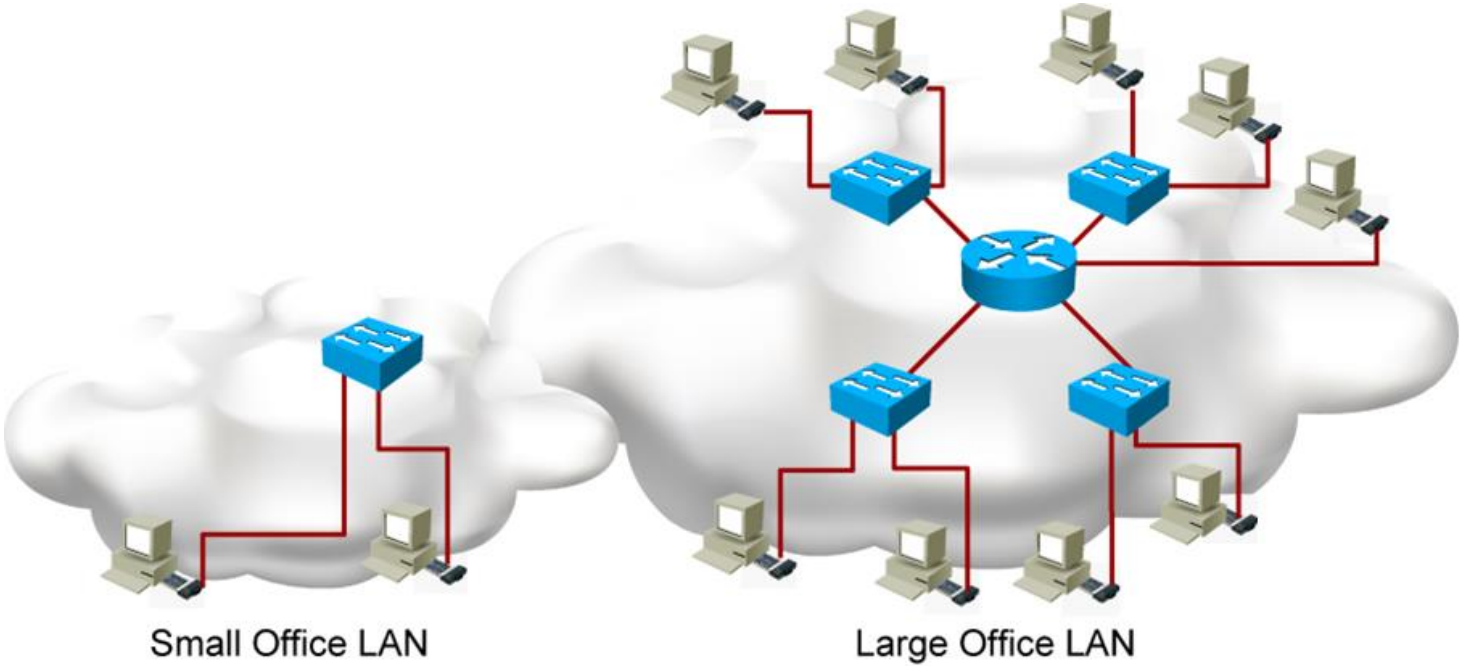
Understanding Ethernet

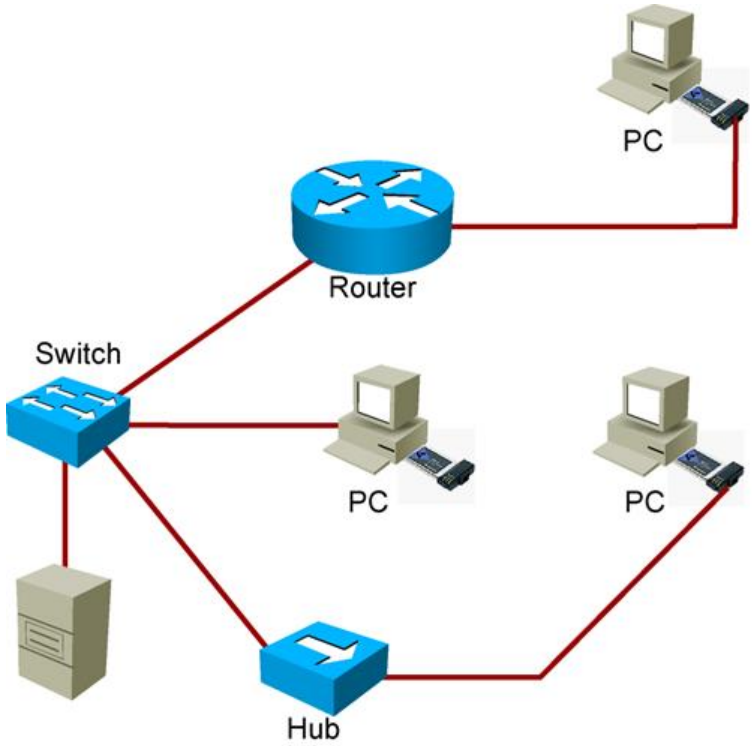
Understanding Ethernet

Local Area Network



LAN Components

- Computers
  - PCs
  - Servers
- Interconnections
  - NICs
  - Media
- Network devices
  - Hubs
  - Switches
  - Routers
- Protocols
  - Ethernet
  - IP
  - ARP
  - DHCP



Functions of a LAN

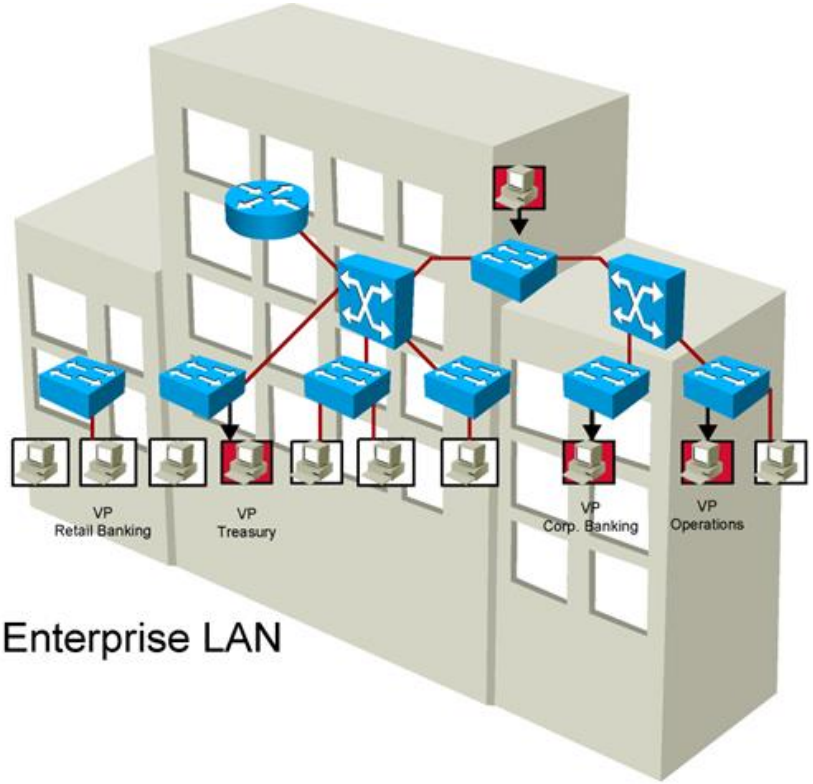
- Data and applications
- Share resources
- Provide communication path to other networks



LAN Sizes



SOHO
LAN



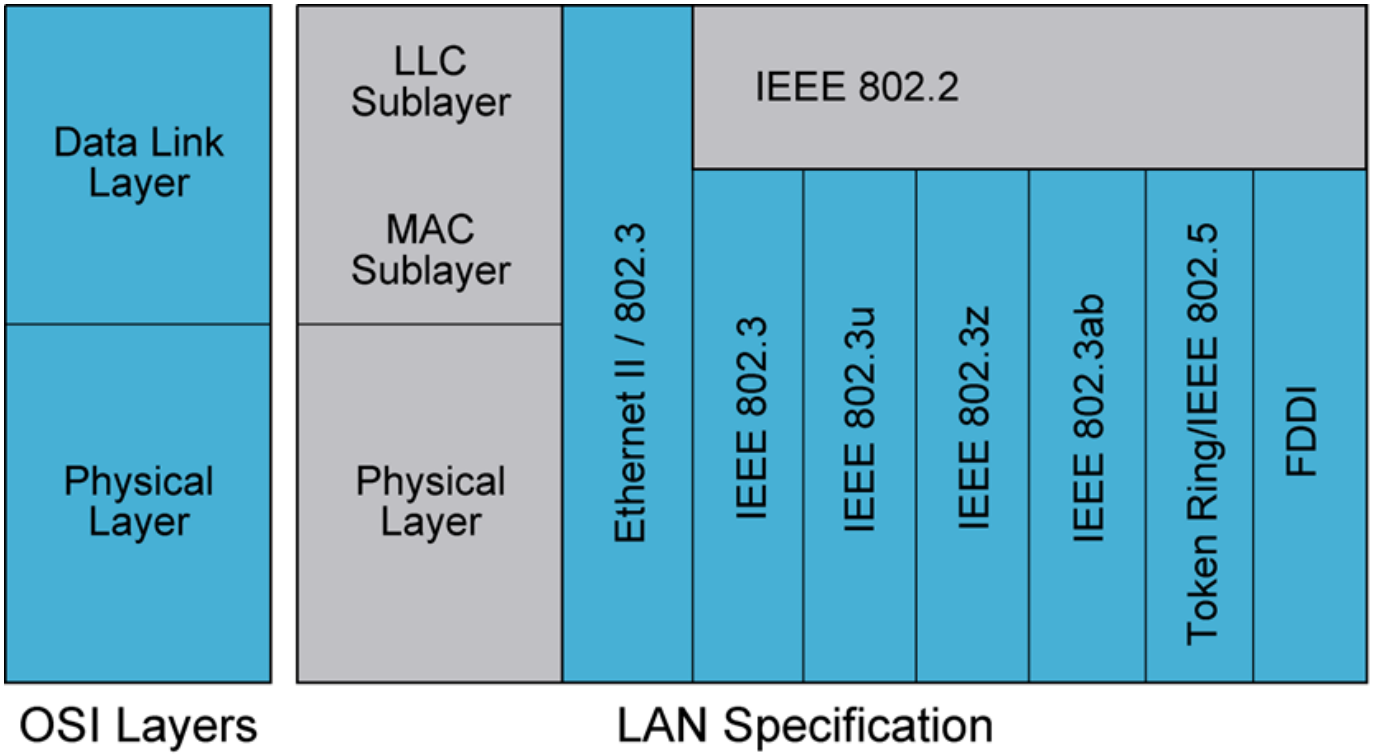
Enterprise LAN

Ethernet Evolution

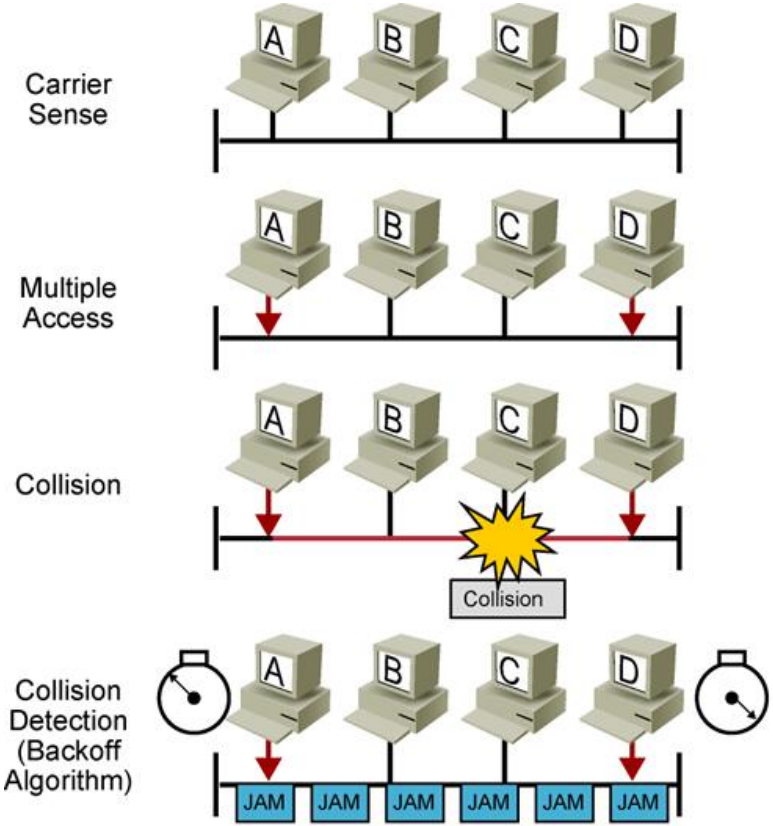
Year	Ethernet Activity
1970	First packet radio network
1973	Ethernet invented at Xerox
1977	U.S. patent no. 4063220 issued
1982	DIX releases 10-Mb/_s Ethernet
1992	First stackable Ethernet hub
2002	IEEE approves 802.3ae; 10 billion bps



LAN Standards

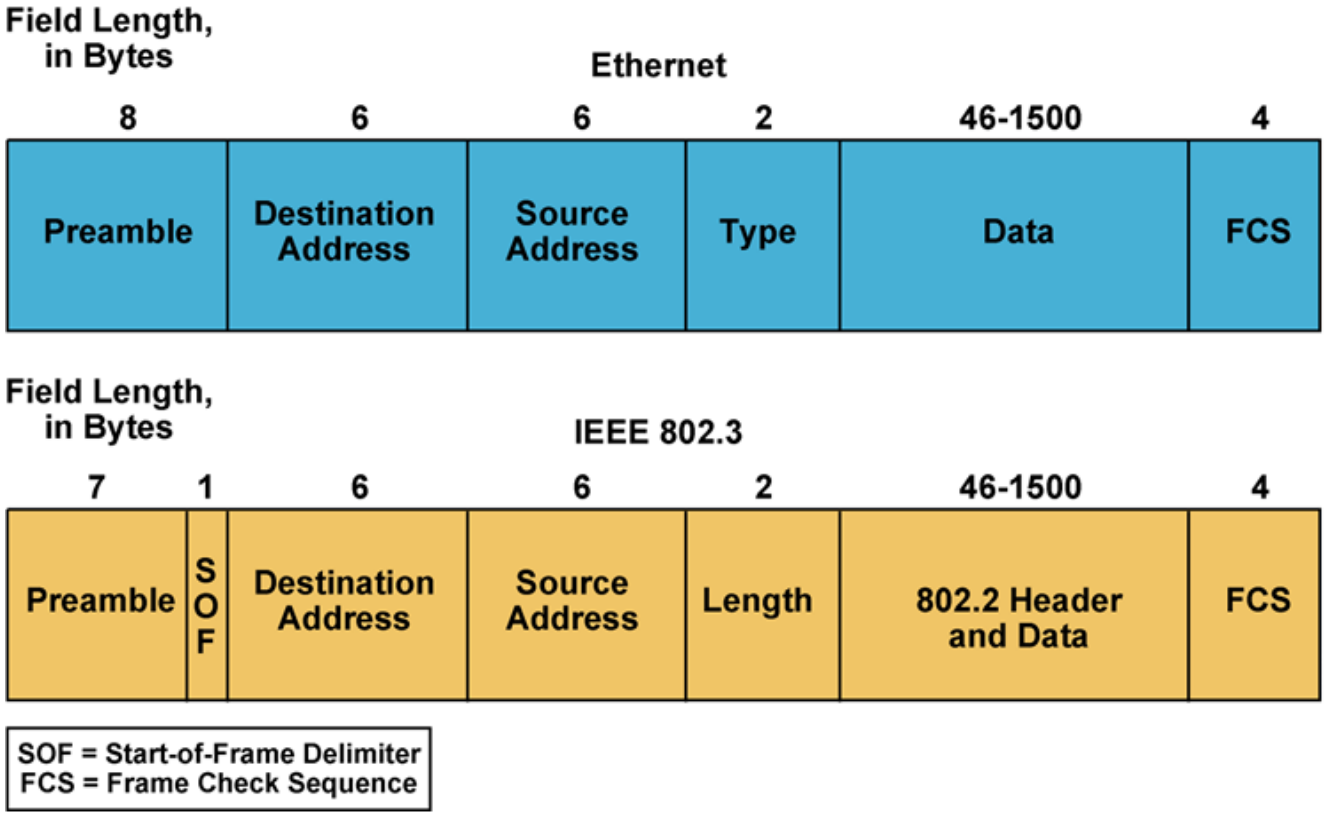


CSMA/CD

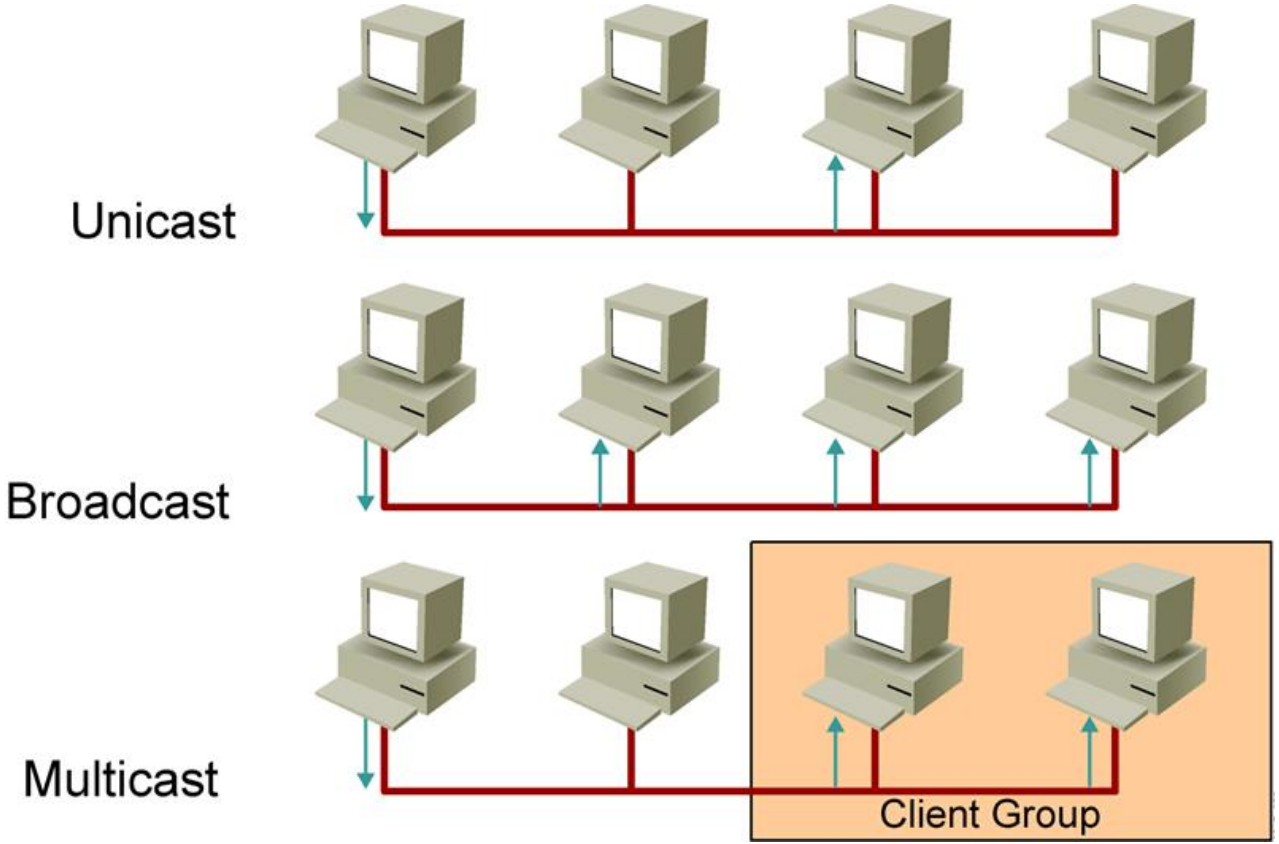


Carrier Sense Multiple Access Collision Detection (CSMA/CD)

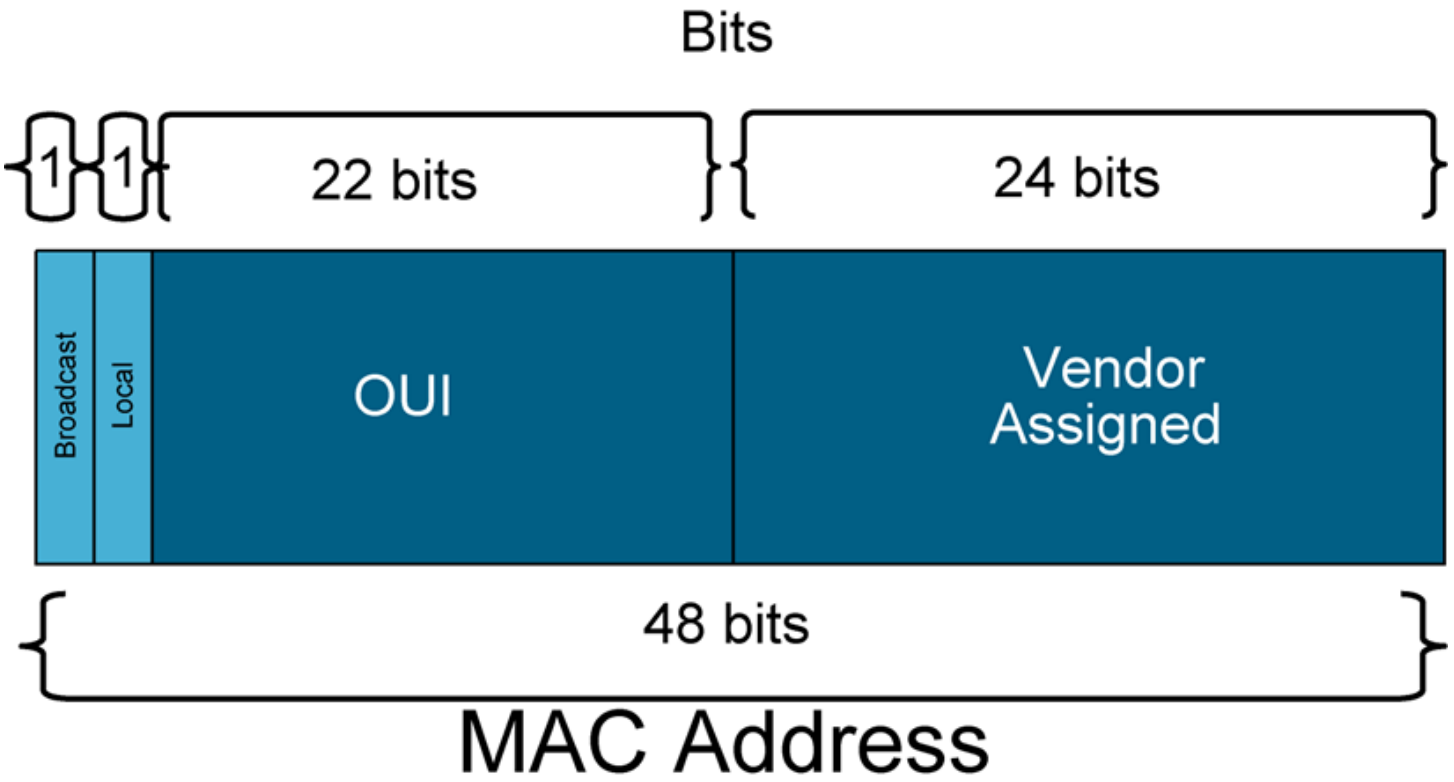
Ethernet Frame Structure



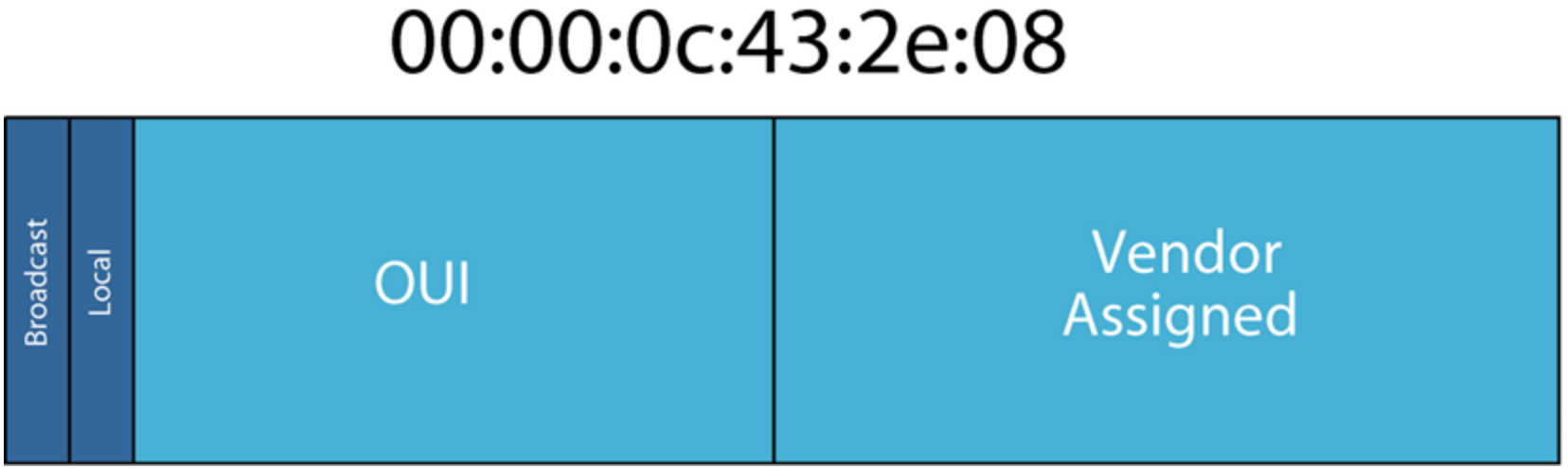
Communicating Within the LAN



MAC Address Components

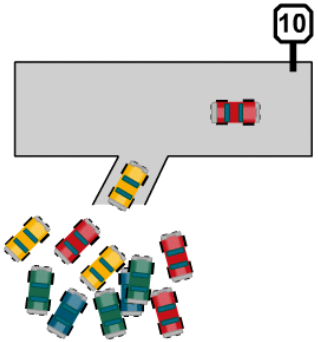


MAC Addresses

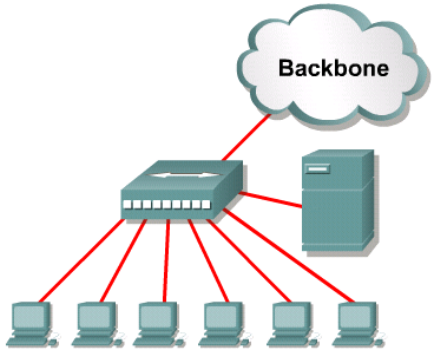


Switch Introduction

Ethernet LAN



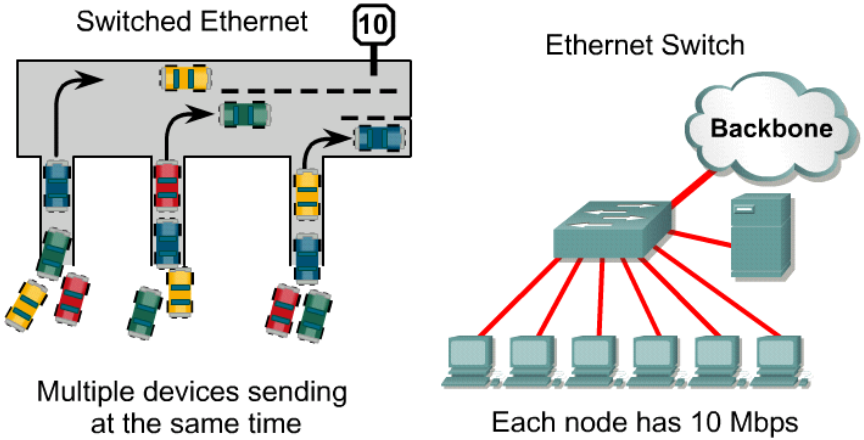
One device sending at a time



Each node shares 10 Mbps

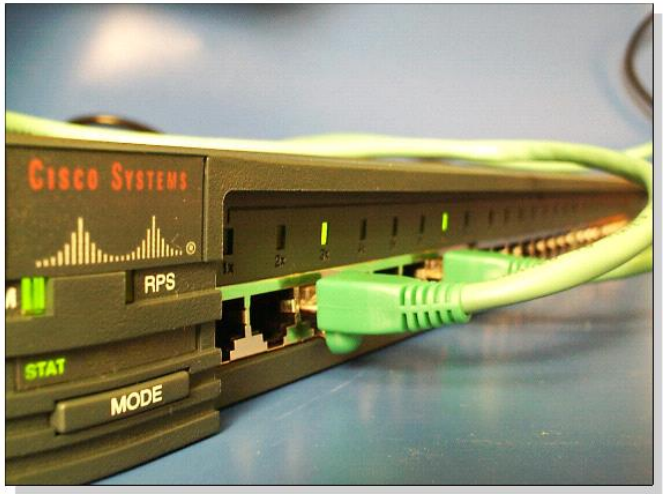
- Ethernet is fundamentally a shared technology where all users on a given LAN segment compete for the same available bandwidth.
- This situation is analogous to a number of cars all trying to access a one-lane road at the same time. Because the road has only one lane, only one car can access it at a time.
- The introduction of hubs into a network resulted in more users competing for the same bandwidth.

Ethernet/802.3 LAN development



- In a pure switched Ethernet LAN, the sending and receiving nodes function as if they are the only nodes on the network.
- When these two nodes establish a link, or virtual circuit, they have access to the maximum available bandwidth. These links provide significantly more throughput than Ethernet LANs connected by hubs.
- This virtual network circuit is established within the switch and exists only when the nodes need to communicate.

Ethernet/802.3 LAN development

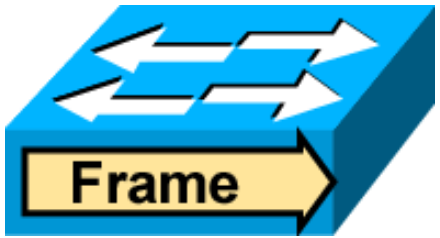


- A switch is a Layer 2 device and may be referred to as a multi-port bridge.
- A switch will segment a LAN into microsegments which decreases the size of collision domains.
- A switch has the intelligence to make forwarding decisions based on MAC addresses contained within transmitted data frames.
- The switch learns the MAC addresses of devices connected to each port and this information is entered into a switching table.

Transmitting Frames

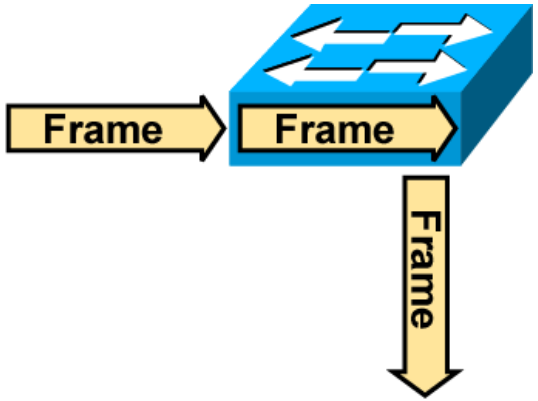
Cut-Through

Switch checks destination address and immediately begins forwarding frame.



Store and Forward

Complete frame is received and checked before forwarding.

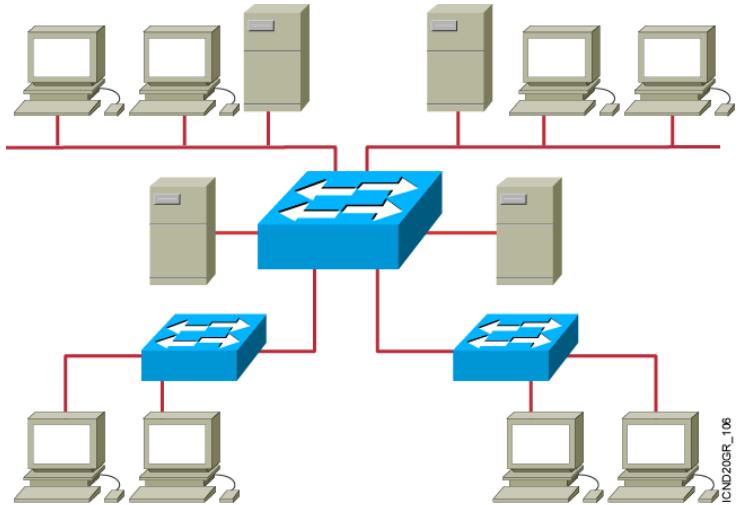


Fragment-Free

Switch checks the first 64 bytes, then immediately begins forwarding frame.

```
SW(config)#    switching-mode
[cut-through | fragment-free | store-and-forward]
```

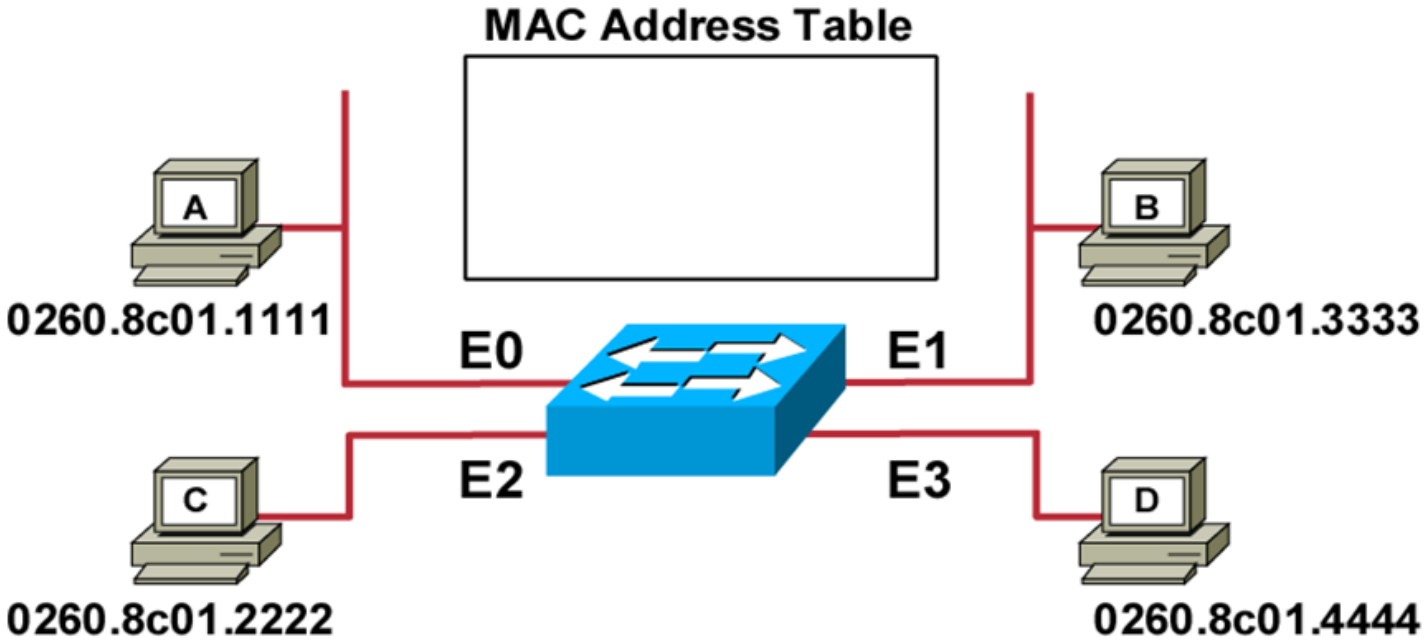
Ethernet Switches and Bridges



- Address learning
- Forward/filter decision
- Loop avoidance

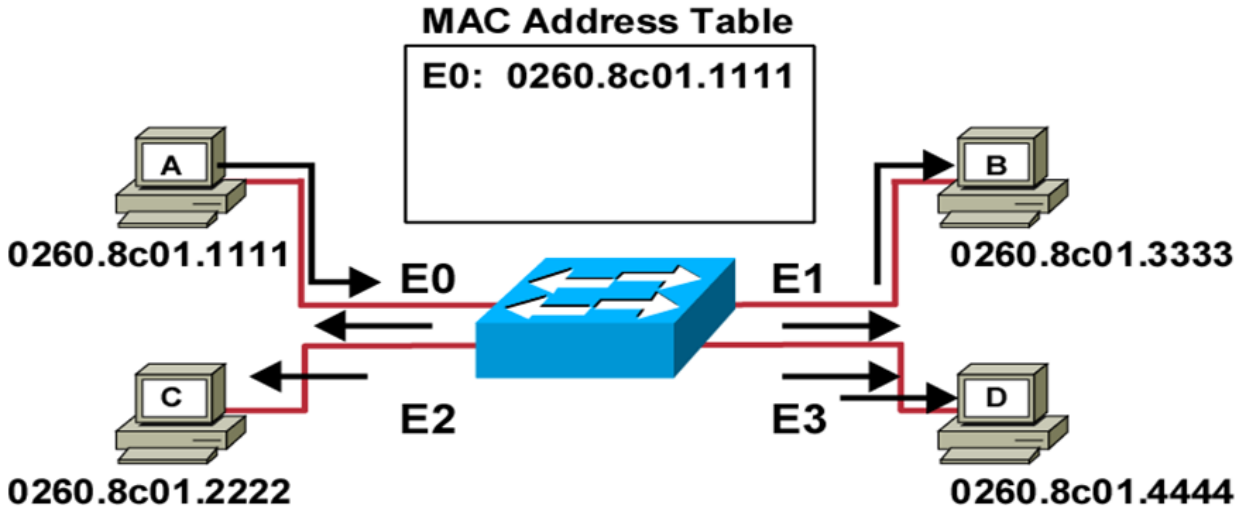


MAC Address Table



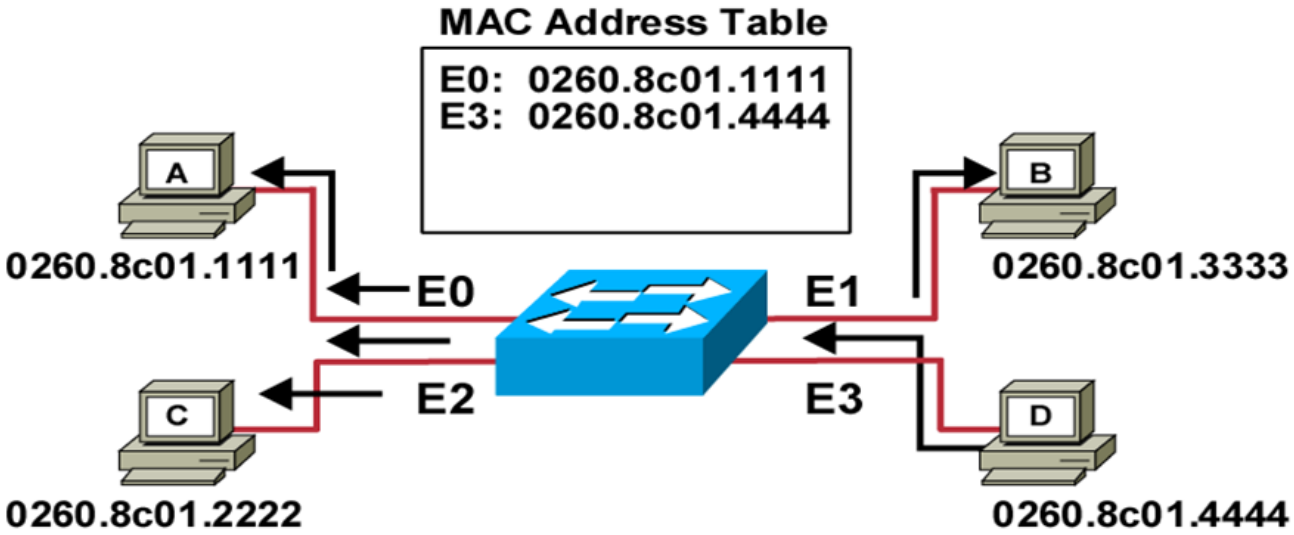
- Initial MAC address table is empty.

Learning Addresses



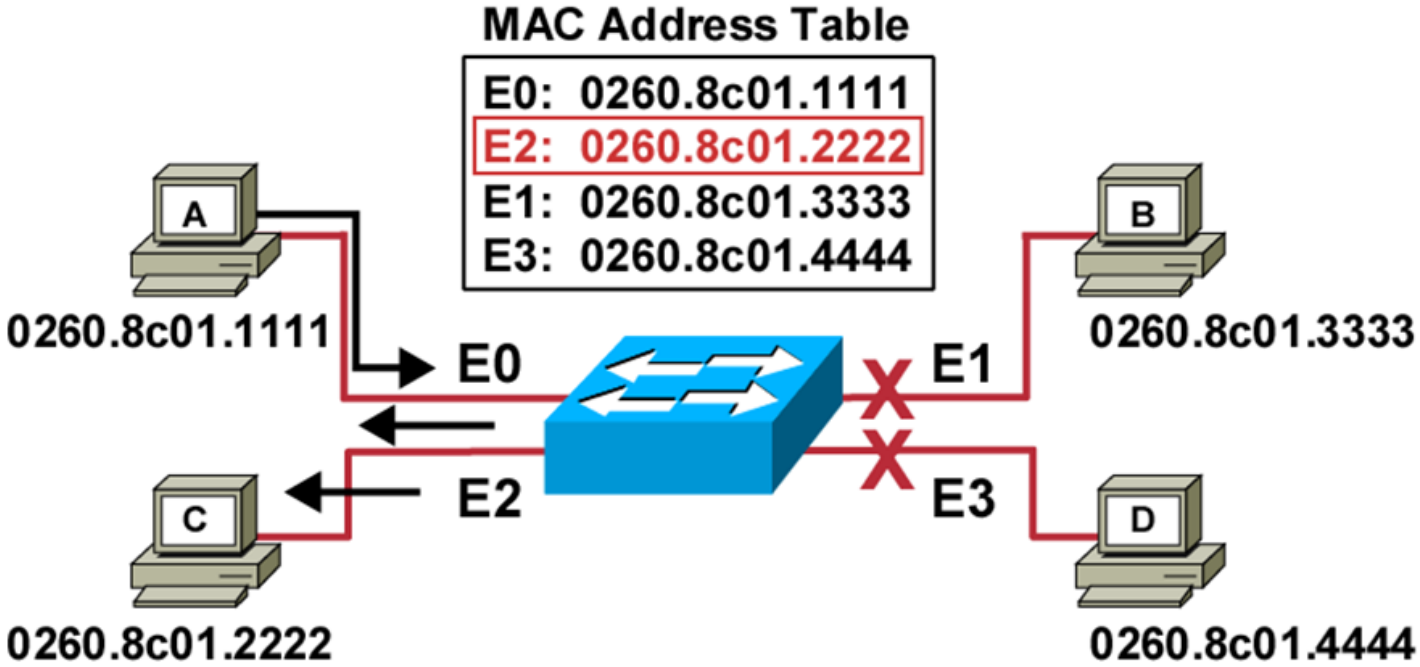
- Station A sends a frame to station C.
- Switch caches the MAC address of station A to port E0 by learning the source address of data frames.
- The frame from station A to station C is flooded out to all ports except port E0 (unknown unicasts are flooded).

Learning Addresses (Cont.)



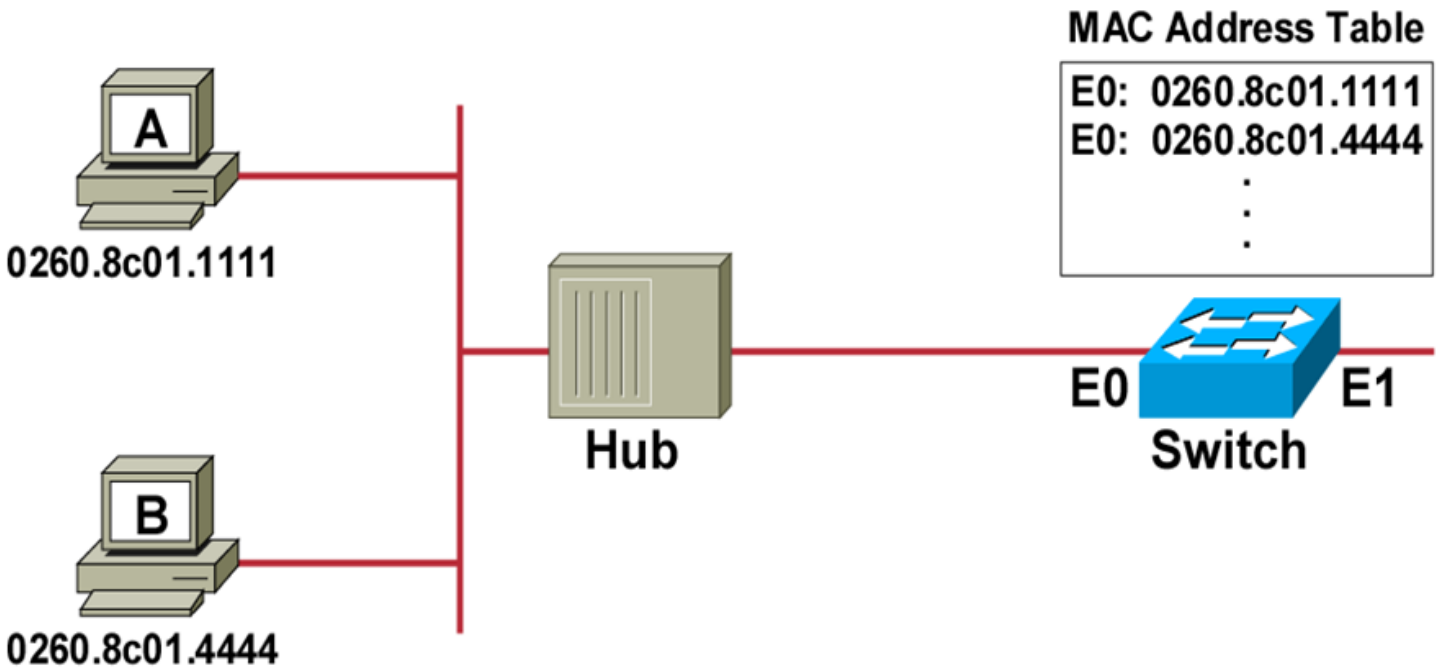
- Station D sends a frame to station C.
- Switch caches the MAC address of station D to port E3 by learning the source address of data frames.
- The frame from station D to station C is flooded out to all ports except port E3 (unknown unicasts are flooded).

Forward Frames



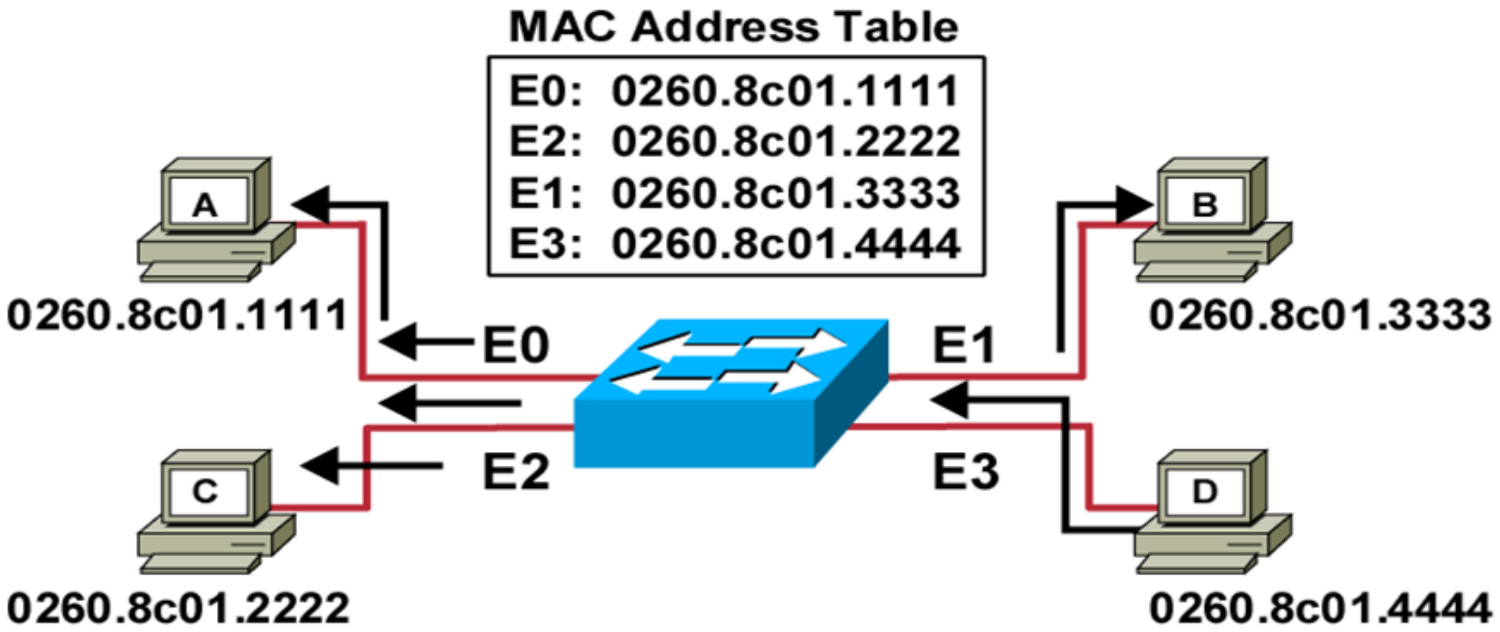
- Station A sends a frame to station C.
- Destination is known; frame is not flooded.

Filtering Frames



- Station A sends a frame to station B.
- The switch has the address for station B in the MAC address table.

Broadcast and Multicast Frames



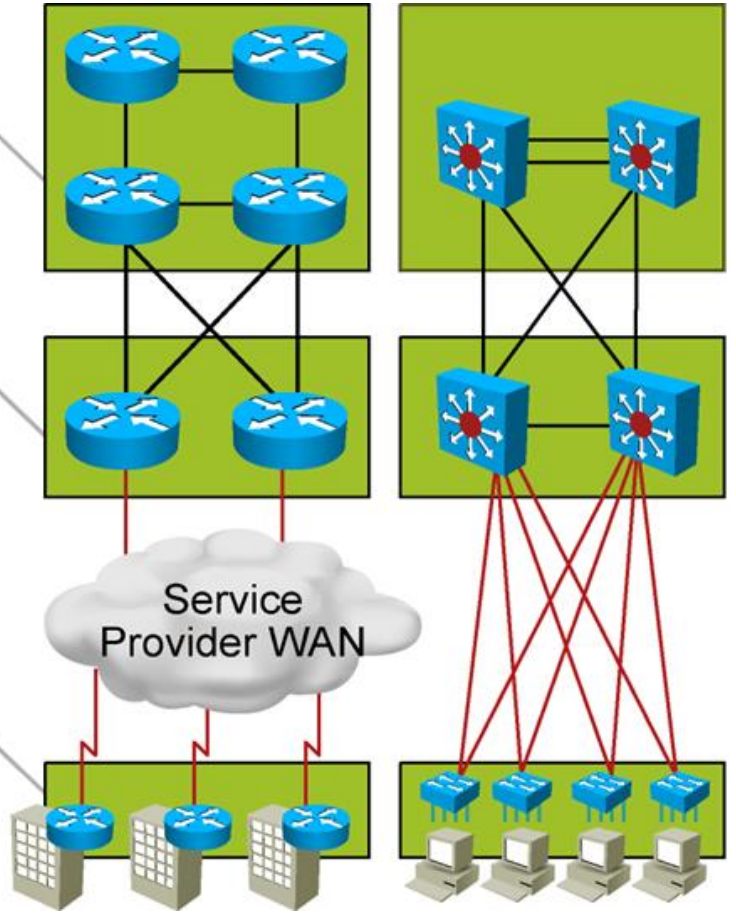
- Station D sends a broadcast or multicast frame.
- Broadcast and multicast frames are flooded to all ports other than the originating port.

The Hierarchy of Connectivity

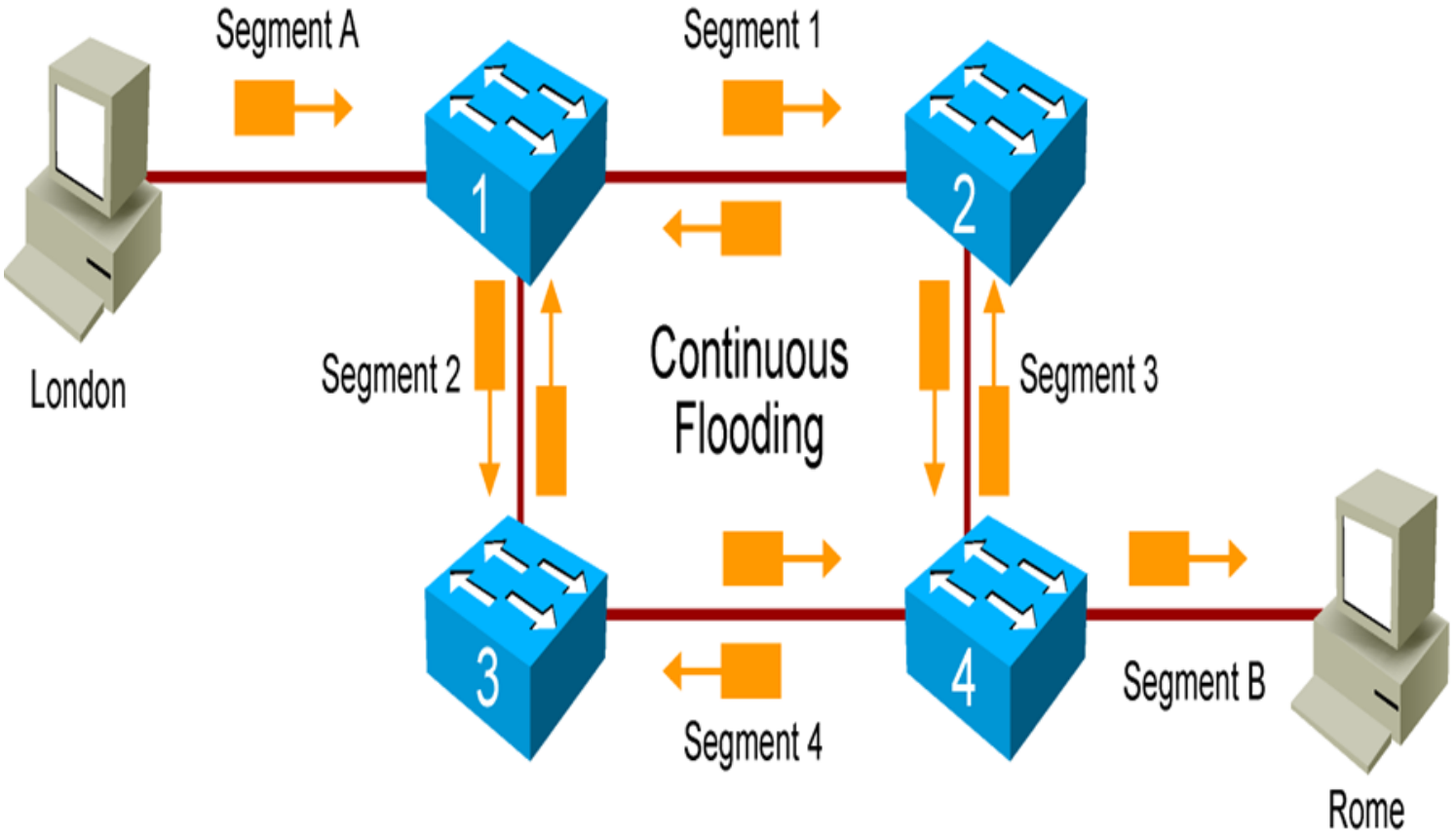
Core layer: Provides optimal transport between core routers and distribution sites

Distribution layer: Provides policy-based connectivity, ? peer reduction, and aggregation

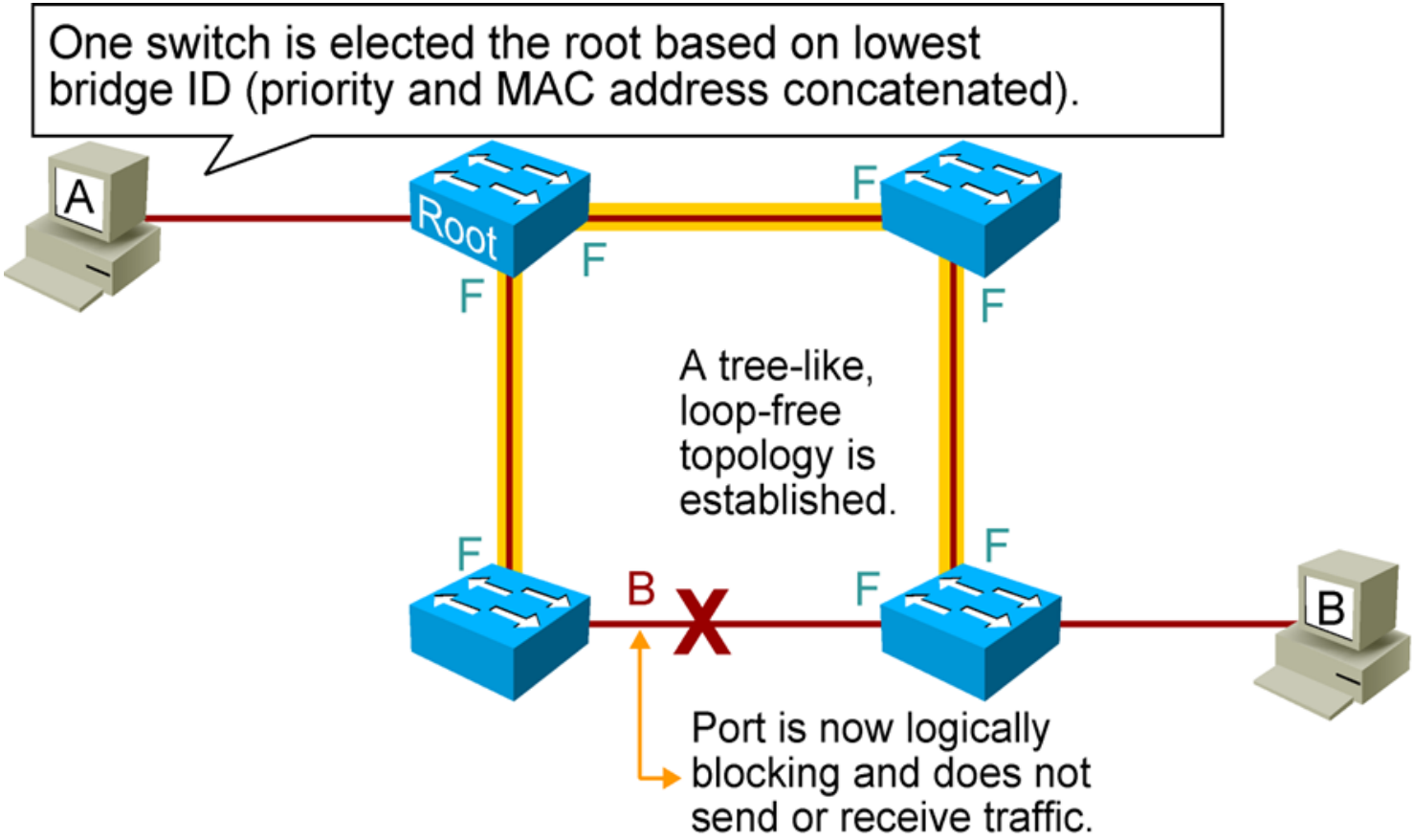
Access layer: Provides common group access to the internetworking environment

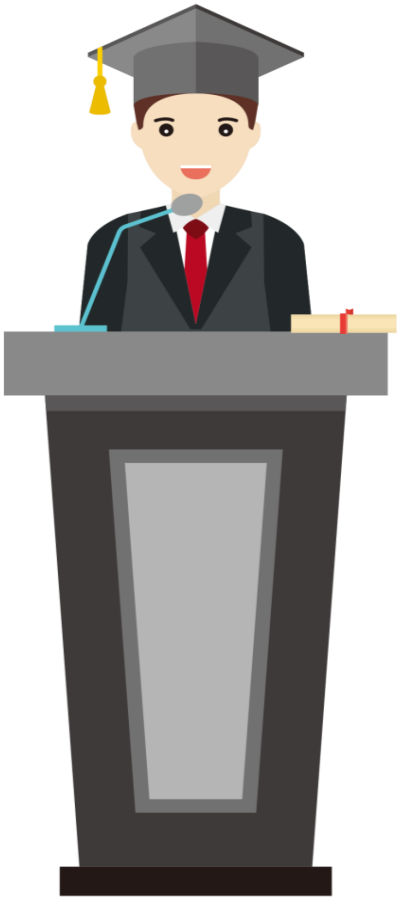


Loops


Segment A
Segment 1
1
2
London
Segment 2
Continuous
Flooding
Segment 3
3
4
Segment B
Segment 4
Rome

Spanning Tree Protocol





THANK YOU!

